Fiscal Year:	FY 2010	**Task Last Updated:**	FY 01/05/2010
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Fractional gravity studies of sensory mechanisms in root phototropism–a reflight experiment (TROPI-2)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
Organization Name:	University of North Carolina-Greensboro		
PI Address 1:	College of Arts & Sciences, 105 Foust Building		
PI Address 2:	PO Box 26170		
PI Web Page:			
City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	04/01/2009	**End Date:**	03/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS STS-130 NOTE: New end date is 3/31/2012 per NSSC info (Ed., 2/19/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University) Correll, Melanie (University of Florida)		
Grant/Contract No.:	NNX09AF11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of the plant Arabidopsis thaliana. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. Intermediate or fractional gravity studies (less than the nominal 1g on Earth) were planned to further study the interaction between tropisms. TROPI-1 was successfully performed on the ISS during late 2006. We obtained data on seedlings grown in microgravity and from the 1g control. However, one problem encountered during TROPI-1 was low seed germination due to long storage periods (8 months) in flight hardware. Thus, the originally proposed fractional gravity studies were not performed. TROPI-2 provides an opportunity to regain the results from these important fractional gravity experiments. The proposed experiments will provide a better understanding of how plants integrate sensory input from multiple light and gravity perception systems. This information is important for growing plants on long-term space missions as part of life support systems. The fractional gravity studies contain 0.16g (Moon) and 0.38g (Mars) treatments, so information to be obtained is relevant to the exploration agenda at NASA. We plan to use the European Modular Cultivation System (EMCS) and are working toward a launch date of February 2010 on STS-130. Mission operations on the ISS are planned for February 2010.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	1. Seed storage test at ARC. Our hypothesis was that seed germination decreases over time when seeds are stored in EUE. This study was to test seed germination after one, three, and five months storage in flight EMCS ECs and EUE. The PI team concluded that seeds can be stored in the EUE for 1-3 months without significant deleterious effects. However, seed germination begins a precipitous drop after 3 months of storage in EUE. Thus, the TROPI-2 space experiments should be initiated with no longer than 3 months storage of seeds in hardware. 2. Schedule test at N-USOC. In order to test spaceflight procedures, software, and hardware, the PI team traveled to the Norwegian User Support and Operations Centre (N-USOC) in Trondheim, Norway in June 2009. This facility is the responsible center for the EMCS, an experiment facility located onboard the ISS, and TROPI-2 will be conducted on the EMCS. 3. OVT (Operations Verification Test) at N-USOC. This activity also was performed in Trondheim, Norway in September 2009. We tested the complete time line of the experiment and perfumed the experiment as it will be done during the spaceflight. Thus, we feel confident about the success of TROPI-2. 4. Presentation of results at the 2009 ASGSB meeting. In November 2009, the PI and his team gave several presentations (see the end of this report) on the EMCS project and related ground-based research at the ASGSB meeting in Raleigh, North Carolina. This is an important international meeting for researchers with spaceflight experiments. 5. Preparation for STS-130 launch at ARC. We had a complete flight build of 18 EC's in December 2009. This build was based on a launch date for STS-130 in February 2010. The post-doc and ARC staff performed the final hardware build under careful QA supervision. The pressure seal on the EUE in EC's (experimental containers) was tested. The PI traveled to ARC to approve of the completion of the EUE hardware build. 6. Analysis of frozen samples at University of Florida. One of the goals of this project is to study gene expression as the result of light and gravity-treatments of plants in space. As a first step in this process, RNA needs to be isolated from the frozen plant samples. We are in the process of analysis of RNA and will shortly begin microarray experiments. 7. Throughout the funding year. There were numerous telecons among the PI team at Miami University and the ARC team. Detailed minutes from each telecon are available upon request.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Kumar P, Edemann RE, Kiss JZ. "Studies in preparation of a spaceflight experiment to investigate tropisms." Plant Biology 2009 (Annual Meeting of the American Society of Plant Biologists), Honolulu, HI, July 18-22, 2009. Plant Biology 2009 (Annual Meeting of the American Society of Plant Biologists), Honolulu, HI, July 18-22, 2009. Abstract Book. , Jul-2009
Abstracts for Journals and Proceedings	Millar K, Correll MJ, Edelmann RE, Mullen JL, Hangarter RP, Kiss JZ. "New insights into phototropism from experiments in microgravity." Plant Biology 2009 (Annual Meeting of the American Society of Plant Biologists), Honolulu, HI, July 18-22, 2009. Plant Biology 2009 (Annual Meeting of the American Society of Plant Biologists), Honolulu, HI, July 18-22, 2009. Abstract Book. , Jul-2009
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Articles in Peer-reviewed Journals	Kiss JZ, Kumar P, Millar KD, Edelmann RE, Correll MJ. "Operations of a spaceflight experiment to investigate plant tropisms." Adv Space Res. 2009 Oct 15;44(8):879-86. http://dx.doi.org/10.1016/j.asr.2009.06.007 , Oct-2009
Books/Book Chapters	Molas ML, Kiss JZ. "Chapter 1. Phototropism and Gravitropism in Plants." in "Advances in Botanical Research." Ed. J.-C. Kader, M. Delseny. , New York : Academic Press, 2009. p. 1-34. http://dx.doi.org/10.1016/S0065-2296(08)00601-0 , Jan-2009